' D W D $ O J R U L W K P V

5 H SHWLWLYH DQG ' HVLJQ 3DWWHUQV IRU 6FDODEOH 8 S8 VLQJ 3\ 6SDUN

0 DKP RXG 3DUVDQ

'DWD $OJRULWKPV ZLWK 6SDUN

E\0DKPR3XDGUVLDQ

&RS\ULJKW ‹ 0DKPRXG 3DUVLDQ $OO

3UL QWHG LQ WKH 8QLWHG 6WDWHV RI $P

3XEOLV2¶H5GHLE0O\ 0HGLD*U,QF\HQVWHLQ +L
6HEDVWRSRO &$

2¶5HLOO\ ERRNV PD\ EH SXUFKDVHG IRU
SURPRWLRQDO XVH 2QOLQH HGLWLRQV
KWWS FRUPH)RUOPRUH LQIRUPDWLRQ FRQ
FRUSRUDWH LQGLVSHQVDWLRQV R
FRUSRUHDLWFH#R U

$FTXLVLWLRQHQWV(GLWRDE HUPDQ

'HYHORSPHQWO(GLWDR3UR WWHU

3URGXFWL&RQ(GLWRSUKHU)DXFKHU

&RS\HG5LWFRHO +HDG

3URRIUHDGHU %LOOLQJ

,QGH3HRWR PDF ,QGH[LQJ //&

,QWHULRU'HVLJQ)QHWDR

&RYHU 'HVLJQHU0RQWJRPHU\

,OOXVWUDWRU'HO O HD

$SUIL)LUVW (GLWLRQ

5HYLVLRQ +LVWRU\ IRU WKH)LUVW (GLW

-)LUVW 5HOHDVH

6HHWWS FRUHDWDORJ HUUDWD FRP"LVEQ
GHWDLOV

7KH 2¶5HLOO\ ORJR LV D UHJLVWHUHG WUDGHPDUN
$OJRULWKPV ZLWK FRSYHUNLPDJH DQG UHWUDGHPDUNV RI 2¶5HLOO\ 0HGLD ,QF

7KH YLHZV H[SUHVVHG LQ WKLV ZRUN DUH WKRVH RI WKH DXWKRU V DQG GR QRW UHSUHVHQW WKRVH RI 2¶5HLOO\ 0HGLD ,QF 7KH LQIRUPDWLRQ DQG LQVWUXFWLRQV SURYLGHG LQ WKLV ZRUN DUH EHOLHYHG WR EH DFFXUDWH DQG UHOLDEOH KRZHYHU WKH SXEOLVKHU DQG WKH DXWKRU PDNH QR UHSUHVHQWDWLRQV DERXW WKH DFFXUDF\ RU FRPSOHWHQHVV RI DQ\ FRQWHQWV RI WKLV ZRUN DQG VSHFLILFDOO\ GLVFODLP DOO UHVSRQVLELOLW\ IRU HUURUV RU RPLVVLRQV LQFOXGLQJ ZLWKRXW OLPLWDWLRQ UHVSRQVLELOLW\ IRU GDPDJHV DULVLQJ IURP UHOLDQFH RQ DFFXUDF\ RU FRPSOHWHQHVV RI DQ\ FRQWHQW LQ WKLV ZRUN 8VH RI WKH LQIRUPDWLRQ DQG WKH GVLZQ \RXU RZQ ULVN ,I DQ\ FRGH VDPSOHV RU RWKHU WHFKQRORJLHV LQ WKLV ZRUN DUH VXEMHFW WR WKH RSHQ 6RXUFH RU SURSULHWDU\ OLFHQVHV JRYHUQLQJ WKHLU XVH \RX PXVW FRPSO\ ZLWK VXFK OLFHQVHV DQG RU

> / 6 , @

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' $7$)5$0(5($'(5 $1' ' $7$)5$0(:5,7(5
6SDVUNW)UDPH5HFDGHV LV DQ LQWHUIDFH WR UHDG
VRXUFHV²VXFK DVQGH6V1 &69HV 3DUTXHW DQG 25&
'DWDEDVH &RQQHFWLYLW\ -'%& FRPSOLDQW GDWD
'DWD)UDPH:UWHV LV DQ LQWHUIDFH WR ZULWH D
VRXUFH
```

2U ZH FDQ6SXDHNW&KRQWH[W

```
!!! LQSXWBSDWKFRGH FKDS VDPSOH IDVWD
!!! /HW VSDUN EH DQ LQVWDQFH RI 6SDUN6HVVLRQ
!!! VF VSDUNSDUN&RQWH[W
!!! UHFRUGVBUGGWH[W)LDHSXWBSDWK
'HJLQH WKH LQSXW SDWK
&UHDWH DQ6SDUNSRQWH[W
8VH 6SDUN&RQWH[W DG LQSXW>6WULQ
```

7KH VHFRQG RSWLRQ LV SUHIHUDEOHW ET
RQH ZRUNVWRHLVELXHQWWE¶HFDXVH LW ILU
WKHQ FRQYHUWV LW WR DQ 5'' DQG HYH
WUDQVIRUPDWLRQ

1H[W ZH¶OO H[DPLQH WKH FRQWHQWV RI DV6WULQJV GHQXRWHGHFEHQW!

```
!!! UHFRUGVBFRGGHFW
>
 X!VHT
 XF*7$DFFDDWDDDDDDDFDDJFWWDDFFWDDWWF
 X!VHT
 XDJFWWDJ777**DWFWJJFFJJJJ
 X!VHT
 XJFJJDWWWDFWF&&&&&$$$$11DJJJJDJDJFFFDJDWD

 XJDDWWFJFDFFD
 X$$7$$$$&&7&$&&&$7
 XDJDJFFFDJDDWWWDFWF&&&
 X!VHT
 XJFJJDWWWDFWFDJJJJDJDJFFFDJ**DWDDDWJJDJWFW
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X JDDWWFJFDFFD
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7,3

7KH '' FROOHFWLRQ LV XVHG KHUH WR UHWULHYH WKH
REMHFWV DQG VHOHFWV RQH RU PRUH 5''V \RX VKRXOG F
FROOHFWLRQ PLJKW FDXVH 220 HUURUV DV ZHOO
SHQDOWLHV YL1HZLOWHPHQLWWR I DQ 5"" WRXNHD\ XVH

6WHS 'HILQH D 0DSSHU)XQFWLRQ

7R PDS 5'' HOHPHQWV LQWR RBOHWWHUHH¶SO
WR GHILQH D 3\WKRQ IXQFWLRQ WKDW
WUDQVIRUPV D WR PDQ\ WUDQVIR
QHZ 5'' E\ ILUVW DSSO\LQJ D IXQFWLRQ \
DQG WKHQ IODWWHQLQJ WKH UHVXOWV
WR IRUP D VLQJOH 5'' 7KH
WKDW ZLOO EH IODWWHQHG
,QIRUPDWLRQ ZLOO ZULWH WKLV DV

&UHDWH DQ LWHUDEOH OLVW

VLQJOHB5''BHOHPHQW 9 !9>@

)ODWWHQ WKH OLVW LQWR

>9 9 9@ ! 9 9 9

)RU WKLV VROXWLRQ
WKDW DFFHSWV DQ 5'' HOHP
6WULQJ UHWXUQV

JLYHQ WKH LQSXW LV D GHVFULSWLRQ UHFRU
SDLUV UHFDOO WKDW ZH¶UH FRQYHUWL

D
D
W
W
J

,I WKH LQSXW LV D GHVFULSWLRQ UHFRU
EHJLQV VZLWK WKH ZRUG HPSW\ WKLV ZLOO HQDE
QXPEHU RI VHTXHQFHV DV ZHOO ,I WKH
WRNHQL]H LW E\ FKDUDFWHUV DQG WKHQ
GQDBOHWWHU HPQW BOHWWHULQDZOUHWXU
WKHVH SDLUV 7KH IXQFWLRQ GHILQLWLR
SULQWV DWHPHQWV IRU GHEXJJLQJ SXUSR
HQYLURQPHQW WKHVH VKRXOG EH UHPRY
SHQDOWLHV

```
3DUDPHWHU IDVWDBUHFRUG 6WULQJ D VLQJOH )$

2XWSXW D OLVW RI NH\ YDOXH SDLUV ZKHUH NH
      LV D GQDBOHWWHU DQG YDOXH LV D IUHTXHQ

GHISURFHVVB)$67$BUDVWDBUHFRUG
    NH\BYDOXHBOLVW

    LI IDVWDBUHFRUGBWVZLWK
        ] FRXQWV WKH QXPEHU RI )$67$ VHTXHQFHV
        NH\BYDOXHBDSSHQG
    HOVH
        FKDUVIDVWDBUHFRUGBORZHU
        IRUF LQFKDUV
            NH\BYDOXHBDSSHQG

    SULQWNH\BYDOXHBOLVW
    UHWXUQNH\BYDOXHBOLVW
HQG GHI
```

&UHDWH DQ HPSW\ OLVW WR KROZKHLFKDZLH
RXWSXW IURP WKLV IXQFWLRQ

$SSHQG WR WKH OLVW
$SSHQG WR WKH GLVW Z1K$HOHWWHU
IRU GHEXJJLQJ SXUSRVHV RQO\
5HWXUQ D OLVW RI SNDLUV ZKLFK ZLOO
IODW0DS WUDQVIRUPDWLRQ

1RZ ZH ZLOO XVH WKLV IXQFWLRQ WR
WKUHH FRUGVB5''>6WULQJ@ MXVW FUHDWHG

!!! UHF UHIHUV WR DQ HOHPHQW RI UHFRUGVBUGG
!!! /DPEGD LV D QRWDWLRQ WKDW GHILQHV LQSXW
!!! LQSXW UHF DV D UHFRUGVBUGG HOHPHQW
!!! RXWSXW UHVXOW RI SURFHVVB)67BUHFRUG
!!! SDLUVBUGG UHFRUGVB0DS SDLUVBUHF
SURFHVVB)67BUHF

7KH VRXUFH5''>6WULQJ@
:H XVH D ODPEGD H[SUHVVLRQ WR
UHFRUGVBUGG 5''>6WULQJ@ 5''>6WULQJ@ DQ
5''> 6WULQJ ,QWHJHU @

$OWHUQDWLYHO\ LW DV IROORZV
H[SUHVVLRQ

!!! SDLUVBUGG UHFRUGVB0DS SURFHVVB)67BUHFRUG

)RU H[DPSOH LIUHF FRUGVBEHRQGVWRLQ WKH
DVJDWWFWKHQ LW ZLOO EH IODWWHQHG
SDLUV

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D
D
W
W
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J

,I DQ HOHPHQW RI GVBUHF RQWDLQV WKHQ LW ZLO
LQWR WKH IROORZLQJ SDLU\ UHF]OWRWKLQ

WRWDO QXPEHU RI '1$ VHTXHQFHV IRU D

]

6WHS)LQG WKH)UHTXHQFLHV RI '

SDLUVBUQGZ FRQWDLQVYDOXHW SDLUV\ZKH
'1$ OHWWHU DQG WKH YDOXH[WLVWLQJ SUH
UHGXFH%\.HUDQVIRUSDLUVRQG WKH
IUHTXHQFLHV IRU DOO '1$ OHWWHUV

7KHUHGXFH%\.HUDQVIRUJHQHUDWHV PHU
NH\ XVLQJ DQ DVVRFLDWLYH DQG FRPPX
DGGLWLRQ DV RXU 7UHGXFHIXQFWLRQ ZH
VLPSO\ WDNH[XPXOOLQJ WRJHWKHU XH
WKHQH[W YDOXH,QWKHLQSXW ILOH\HV D
5'' . . . DQG WKHQ V
UHGXFH%\.HUDQVIRUPDWLRQ ZLOO WUDQ
VLQJOH SDLUEHFDXVH
ZHUH VWRUHG RQ WZR SDUWLWLRQV HD
DQG LQGHSHQGHQWO\

3DUWLWLRQ ^

. . ! . .
5HVXOW RI 3DUWLWLRQ .

3DUWLWLRQ ^

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. . ! . .

1H[W ZH PDS WUDQVIRUPDWLRQ WR D WKDW UHPRYHV DOO SXQFWXDWLRQ U VSDFH DQG FRQYHUWV DOO OHWWHU\ VLPSOH 3\WKRQ IXQFWLRQ

```
LPSRUWUHJH
GHIQRBSXQFWXDWLRQBUHPRYHUGBVWU
    H[FOXGHVWWULQSXQFWXDWLRQ
    W    MRLQKIRUFKLQHFRUGBLVWFKRWQH[FOXG
    WULPPHGHVXE?V      W
    UHWXUQULPPHG
 HQG GHI

UGGBFOHDQUHGGPDSQRBSXQFWXDWLRQ
UGGBFOHDFRQHGHFW
>
  IR[ UDQ  IDVW
  IR[ MXPSHG RI IHQFH
@
```

: H WKHQ IDSWO\DSWUDQVIRUUGWERQDWQF WRNHQL]LQJ WKH HOHPHQWV RI WKLV

```
IODWWHQHGGBFOHDQDWDSPEGYDY VSOLW
IODWWHQRHOFW
> IR[  UDQ  IDVW  IR[  MXPSHGRI  IHQF@
```

)LQDOWKHWHUWUDQVIRUPDWLRQ GURS\ IODWWH5'HGNHHSLQJ RQO\ HOHPHQWV

```
ILQDOBUGGDWWHQHGWHUUDPEGZDOHQ !
ILQDOBUFRG OHFW
> IR[  UDQ  IDVW  IR[  MXPSHGIHQF@
```

```
!!! GI SULQW6FKHPD
 ERXQGHWKRGWD)UDPHLQW6FKHPDWD)UDPH
>
 QDPHVWULQJ
 DJH LQW
 VDODUVWULQJ
@
 ([SOLFLW VFKHPD GHJLQLWLRQ
 {QJRUFHDQH[SOLFLWVFKHPD
 ([SOLFLW FROXPQ QDPHV
```

<RX FDQ DOVR UHDG LQ DQ 5''+>'6WUHDP@RQ

```
!!! UGG VSDUNSDUN&RQWH[W)LOH
!!! UGGFROOHFW
>
 XDOH[
 XDGHO
 XDGHO
 XMDQH
 XMDQH
 XDOH[
 XPDU\
 XPDU\
 XPDU\
 XPDU\
@
```

: ULWLQJ +DQGOLQJ)LOHV

3\6SDUN$3, HQDEOHV XV WR VDYH RXU 5''\ ILOHV)LUVWOVRNOEW¶KRZ WR VDYH DQ 5'' L

```
!!! SDLUV> DOH[ DOH[
          MDQH MDQH @
!!! UGG VSDUNSDUN&RQWH[W)LOH
!!! UGGFROOHFW
> DOH[ DOH[ MDQH MDQH @
!!! UGGFRXQW

!!! UGGVDYH$V7H[W)LOH ORFDOKRVW WHVW SDL
```

7KH '' VDYH$V7H[W)LOHPHWKRG ZULWHV W
GDWDVHW DV D WH[W ILOH RU VHW RI W
ILOHV\VWHP +')6 RU DQ\ RWKHU +DRRS
FDOOVRVWHULQJ RQ HDFK HOHPHQW
WH[W LQ WKH ILOH

1H[WVWHSLQ¶LQH ZKDW LV FUHDWHG LQ +'
ILW WKH SDJH

```
  KGIV GIV  OV KGIV   ORFDOKRVW      WHVW SDLUV
)RXQG   LWHPV
 UZ U  U           KGIV   ORFDOKRVW      WHVW SDLU
 UZ U  U           KGIV   ORFDOKRVW      WHVW SDLU
 UZ U  U           KGIV   ORFDOKRVW      WHVW SDLU

 UZ U  U           KGIV   ORFDOKRVW      WHVW SDLU

  KGIV GIV  FDW KGIV   ORFDOKRVW      WHVW SDLUV
 DOH[
 DOH[
 MDQH
 MDQH
```

7KH UHDVRQ ZH UHWULHV LV EHFDXVH WKH
SDUWLWLRQV

```
!!! UGGJHW1XP3DUWLWLRQV
```

,I \RX ZDQW WR SFDUHWLVH DWKLQQ1RX VKRX
5'' SDUWLWLRQ

```
!!! UGGBVLQJVSDUNSDUN&RQWDLQOOHSDLUV
!!! UGGBVLQFRORHOHFW
> DOH[      DOH[      MDQH      MDQH   @
!!! UGGBVLQJHW1XP3DUWLWLRQV

!!!
UGGBVLQVDYH$V7H[W)LKGIV   ORFDOKRVW      WHVW S
H
```

/HW¶H[DPLQH ZKDW LV FUHDWHG LQ WKH

```
  KGIV GIV  OV KGIV   ORFDOKRVW     WHVW SDLUVB
)RXQG  LWHPV
 UZ U  U      KGIV   ORFDOKRVW     WHVW SDLUVBVL
 UZ U  U      KGIV   ORFDOKRVW     WHVW SDLUVBVL

  KGIV GIV  FDW KGIV   ORFDOKRVW     WHVW SDLUV

 DOH[
 DOH[
 MDQH
 MDQH
```

: H FDQ VDYH D 'DWD)UDPHDLQQDWWR D+P)H6: UEL\V

```
!!! SDLUV> DOH[      DOH[
           MDQH      MDQH  @
!!!
!!! SDLUV')VSDUNUHDWH'DWDSDDDEW
!!! SDLUVV)KRZ

_  B _ B _

_DOH_[  _
_DOH_[  _
_MDQH _
_MDQH _


!!! SDLUVZUL WHV KGIV  ORFDOKRVW     WHVW SDLL
```

+HUVHZKDWUH DWHG LQ WKH +')6

```
  KGIV GIV  OV KGIV   ORFDOKRVW     WHVW SDLUVB
)RXQG  LWHPV
 UZ      KGIV   ORFDOKRVW     WHVW SDLUVBGI B68
 UZ      KGIV   ORFDOKRVW     WHVW SDLUVBGI SDU
F   FVY

 UZ      KGIV   ORFDOKRVW     WHVW SDLUVBGI SDU
F   FVY

  KGIV GIV  FDW KGIV   ORFDOKRVW     WHVW SDLUV
```

```
DOH[
DOH[
MDQH
MDQH
```

<RX PD\ VDYH \RXU 'DWD)UDPH WR D 'DWD)UDPH LQ 3DU
H[DPSOH WR VDYH D 'DWD)UDPH LQ 3DU
WHPSODWH

```
GI LV DQ H[LVWLQJ 'DWD)UDPH REMHFW
IRUPDW RSWLRQV DUH FVY SDUTXHW MVRQ
GIZULWHDYH
    WDUJHW SDWK
  IRUPDWSDUTXHW
  PRGHDSSHQG
```

5HDGLQJ DQG:LQJ +')6 6HTXHQFH)LOH

+DGRRSURV WR SHUVLVW DG6HTXHQFHVSLQH
+')6 6HTXHQFH)LOHV FRQVLVW
SDLUV +DGRRSTXHQFHFLHQFHVV DV
RUJ DSDFKH KDGRRS LR 6HTXHQFH)LOH
SURYLGHTXHQFH)LOH :UHTXHQFH)LOH 5HDG
6HTXHQFH)LOH 6FROOWKHV IRU ZLQJ U
UHSHF6WLTXHQFH)LOWKH VWDQGDUG ELQ
IRU +DGRRS ,W VWRUHVDEVOHHYFRXGHV SDLU
VSOLWWLQJ DG6HTXHQFH)LOHV
LQWHUPHGLDWH GDWDSLSHOLQHV LQLQDFWH
HIILFLHQW WKDQ WH[W ILOHV

5HDGLQJ +')6 6HTXHQFH)LOHV

6SDUN VXSSR6HTXHQFH)LOH
6SDUN&RQWH[W VHTXHQFH)LOH)RU H[DP
6HTXHQFH)LOHWHK[WH\V'RXE OH:ULWDEOH
3\WKRQ ZH ZRXOG GR WKH IROORZLQJ

```
  VSDUN DQ LQVWDQFH RI 6SDUN6HVVLRQ
UGG VSDUWSDUN&RQWHHTXHQFH)LOHWK
```

1RWH WKDW XQOLNH ZLWK -DYD RU 6FDO
YDOXH SDLUV WR WKH 6SDUN $3, 6SDUN
7H[WW6RWULQJQRGXEOH:ULWDEOH

:ULWLQJ +')6 6HTXHQFH)LOHV

3\6SDUN5'' NHYDOXH$V6HTXHQFH)LOHWKRG DOORZ
DQ 5'' RI NH\DOXH SDLUV WR EH VDYHG DV DQ H[DPSO
FUHDWH DQ 5'' IURP D 3\WKRQ6HTXHQFH)LOH
DV IROORZV

```
  VSDUN DQ LQVWDQFH RI 6SDUN6HVVLRQ
SDLUV> NH\ NH\
       NH\ NH\ @
UGG VSDUWSDUN&RQWHSDLUV OOHS DLUV
UGGVDYH$V6HTXHQFH)LOH VHTXHQFHILOH
```

:H FDQ WKHQ UHDG6WTXHQFH)LOHV DQG UHDOVHTXHQ
5'' RI NH\DOXH SDLUV

```
  VSDUN DQ LQVWDQFH RI 6SDUN6HVVLRQ
UGG VSDUWSDUN&RQWHHTXHQFH)LOHVHS VHTXHQFHILOH
UGGFROOHFW
> X NH\
  X NH\
  X NH\
  X NH\
@
```

5HDGLQJ DQG :UWLQJ 3DUTXHWVJ LOHV

3DUTXHWDV FROXPQDU GDWD IRUPDW VXSS
V\VWHPV WH,OVI¶GHVFULELQJ PHWDGDWD L
DQG LGHDOQDOUWDFW

6SDUN 64/ SURYLGHV VXSSRUW IRU ERWK
ZKLOH DXWRPDWLFDOO\ SUHVHUYLQJ WK
ZULWLQJ 3DUTXHW ILOHV DOO FROXPQV
IRU FRPSDWLELOLW\ UHDVRQV

)LJXUHLOOXVWUDWHV D ORJLFDO WDEOH
OD\RXWV

SRLQWLQJ WR LW ,Q VKRUW 3DJH5DQN
ZHE DERXW KRZ LPSRUWDQW D SDJH LV

3DJH5DQN &RPSXWDWLRQ

*LYHQ D1 VHW RI SDJHV 3DJH5DQN FDQ EH F
DOJHEUDLFDOO\ DQG LWHUDWLYH PHWKRG FDQ EH

$W DQ LQLWLDO SUREDELOLW\ GL

,% —

$W HDFK WLPH VWHS WKH FRPSXWDW

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,W LV EHOLHYHG WKDW *RRJOH DQG RWI
VFRUHV HDFK WLPH LW FUDZOV WKH ZHE
QXPEHU RI GRFXPHQWV LQ WKH FROOHFW
DSSUR[LPDWLRQ DQG UHDVHV IRU DOO GRF
3DJH5DQN DOJRULWKP UDQNV ZHEVLWHV
OLQNV 7KH TXDOLW\ RI DQ LQFRPLQJ OL
3DJH5DQN HVWLPDWHV WKDW SURYLGHV WKH OLQN

> 127(
>
> 3OHDVH QRWH WKDW WKLV LV DQ H[WUHPHO\ VLPS
> DOJRULWKP *RRJOH DQG RWKHUV ZKR XVH VLPLO
> IDFWRUV VXFK DV WKH QXPEHU RI OLQNV D SDJH
> FDOFXODWLQJ WKH 3DJH5DQN RI D ZHE SDJH

1H[W ,¶OO VKRZ \RX KRZ WKH 3DJH5DQN
IROORZLQJ VLPSOH JUDSK

```
  FDW VLPSOHBJUDSK W[W
$ %
% $
$ '
' $
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6HWWLQJ ZH FDQ ZULWH

```
35 $     G   G  35 %  / %   35 '  / '
35 %     G   G  35 $  / $
35 '     G   G  35 $  / $
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ZKHUH

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/ $
/ %
/ '
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7R FDOFXO3DWHWWHKUHDZHLYQHHOH\G W3R tQ3L5W%D
35 ' :H¶OO LQLWLDOL]H WDKOHQRLWWHKUHDPWWY
35 $ 35 %35 ' XQWLO WKHVH YDOXHV JHR
7KH UHVXOWV7DEUCHH VKRZQ LQ

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,WHUDWLRQ35 $	35 %	35 '
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1RWH WKDW QR PDWWHU ZKDW \RX XVH D
DOJRULWKP VRQ\HRX JHW WKH ZHLJKWV
UHVXOWV ZLWK DQ LQLWLDO]DWLRQ YD

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,WHUDWLRQ3 5 $	3 5 %	3 5 '
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,Q ERWK FDVHV WKH UHVXOWV DUH WKH

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35 $
35 %
35 '
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$V \RX VDZ &KDSWHU 6SDUN SURYLGHV $3,V IRU
3DJH5DQN DOJRULWKP WKURXJK WKH 6SDUN
KHOS \RX XQGHUVWDQG KRZ WKH DOJRUI
FXVWRP 3DJH5DQN VROXWLRQV LQ 3\6SD

&XVWRP 3DJH5DQN LQ 3\6SDUN 8VLQ

,¶OO VWDUW E\ SURYLGLQJ D VLPSOH FX
FRPSOHWH SURJUDP XVLQJ GDWDSDUHH DV
UHSRVLWRU\ JHQHUDWH EDVHG RQ \HDUQBGDWD W

7KLV VROXWLRQ XVHV 6SDUN 5''V WR LW
GRHV QRW XVH *UDSK; RU *UDSK)UDPHV
H[DPSOH ODWHU

,QSXW GDWD IRUPDW

/HW¶V DVVXPH WKDW RXU LQSXW KDV WKH

```
VRXUFH 85/ ,'! ! QHLJKERU 85/ ,'!
```

2XWSXW GDWD IRUPDW

7KH JRDO RI WKH 3DJH5DQN DOJRULWKP

```
85/ ,'! SDJH UDQN YDOXH!
```

7KH UHVXOW ZLOO ORRN VRPHWKLQJ OLN
LWHUDWLRQV

```
VSDUN VXEPLW SDJHUDQN S\ SDJHUDQNBGDWD W[W
KDV UDQN
KDV UDQN
KDV UDQN
KDV UDQN
KDV UDQN
```

3\6SDUN 6ROXWLRQ

2XU FXVWRP VROXWLRQ LQYROYHV WKH

5HDG WKH LQSXW SDWK DQG WKH QXP

```
LQSXWBSDWK DUJY@
QXPBRIBLWHUDWLRQV DUJY@
```

&UHDWH DQ6SDUN6HVVLRQ WKDW XVHV

```
VSDUN6SDUN6HVVLRQ EXLOGHU 2U&UHDWH
```

&UHDWH'DWD)UDPH IURP WKH LQSXW SDWK

```
UHFRUGV VSDUN&RQWH[W)LOHLQSXWBSDWK
```

/RDG DOO WKH 85/V IURP WKH LQSXW

```
GHIFUHDWHBSDLUBUDGBRIBXUOV
    UHFRUGBRIBXUOV    VRXUFH 85/! ! QHLJKEF
  WRNHQVUHFRUGBRIBXUOV
  VRXUFHB85WRNHQ@
  QHLJKERUB85WRNHQ@
  UHWXUQRXUFHB85QHLJKERUB85/
HQG GHI

OLQNVUHFRUGSDSDPEGUDFFUHDWHBSDLU
              GLVWLQFW
              JURXS%\.H\
              FDFKH
```

&UHDWH DVRXUFHB85/ QHLJKERUB85 0DNH VXUH WKHUH DUH QR GXSOLFD)LOG DOO QHLJKERU 85/V &DFKH WKH UHVXOW EHFDXVH LW Z LWHUDWLRQ

7UDQVIRUP 85/ QHLJKERUV LQWR UDQN

```
UDQNVOLQNPVDSODPEGXDOBQHLJKERUVBQHLJKE@RUV
```

UDQNLW 5D'Q 6WULQJ)ORDW @

&DOFXODWH DQG XSGDWH RWKHU 85/V ZH QHHG WZR EDVLF IXQFWLRQV

```
GHIUHFDOFXODWHUDQN
    QHZBUDQNUDQN
    UHWXUQQHZBUDQN
HQG GHI

GHIFRPSXWHBFRQWULXEXWLRQVQN
    FDOFXODWHV 85/ FRQWULEXWLRQV
    WR WKH UDQNV RI RWKHU 85/V
    XUOV XUOVBU>D@Q>@
    UDQN XUOVBU>D@Q>@

    QXPBXUOVHQXUOV
    IRUXUOLQXUOV
        \LHOGXUOUDQNQXPBXUOV
HQG GHI
```

1RZ OHW¶V PHUIRUP WKH LWHUDWLRQV

```
IRULWHUDWLRQVLQDQJHXPBRIBLWHUDWLRQV
    FDOFXODWHV 85/ FRQWULEXWLRQV
    WR WKH UDQNV RI RWKHU 85/V
    FRQWULEXWLRQVQN
        MRLQQDQNV
        IODW0DSSXWHBFRQWULEXWLRQV
```

KRZ HYEHLU GDWD HQJLQHV OLNH +DGRRS DQ
GDWD 6R ZKDW FDQ ZH GR WR PLQLPL]H W
GDWDVHWV" +HUH ,¶OO LQWURGXFH D GHV
QHHG IRU OWHKEQVGKXVRUW SKDVH LQ WRHS0DS
MRLQ

$ PDS VLGH MRLQ LV D SURFHVV ZKHUH WZ
WKDQ XVLQJ WKH DFWXDO MRLQ IXQFWLRQ
PDSSHU DQG,Q DEGXWLRQ WR GHFUHDVLQ
DQG JHQJ LQ WKHDVQGXUHGXFH VWDJHV WK
WKH WDVN SHUPRUPYDQFH

7R KHOS \RX XQGHUVWDQG WKRZWKLW ZRUN
6XSSRVH ZH KDYH WZR WDEOHV3DQG7DQ6
ZDQW WR SHUIRUP D MRLQ EHWZHHQ WKHP

```
P\VT!OXVHWHVWGE
'DWDEDVHKDQJHG

P\VT!OVHOHFWURPHPS

_HPSBL_GHPSBQD_PGHHSWB_LG

_        _DOH[    _        _
_        _WHG     _        _
_        _PDW     _        _
_        _PD[     _        _
_        _MRH     _        _

  URZVQVHW      VHF

P\VT!OVHOHFWURPGHSW

_GHSWB_LGHSWBQD_PGHHSWBORFDWLRQ

_        _$&&28172*1(:<25.1<  _
_        _5(6($5&+ _'$//$6 7;  _
_        _6$/(6    _&+,&$*2,/  _
_        _23(5$7,21_6%267210$  _
_        _0$5.(7,1*_6XQQ\YDOH_
_        _62)7:$5( _6WDQIRU&G$ _

  URZVQVHW      VHF
```

1H[W ZH MRLQ WZR1 W(5E-2H1VR QXGWHKQW BDLG\

```
P\VT!OVHOHHWHPSBLGHPSBQDPHGHSWBLGGHSWBQDPH
GGHSWBORFDWLRQ
        IURHPSH GHSW
            ZKHUHGHSWBLGGHSWBLG

_HPSBLGHPSBQD_PGHSWB_LGHSWBQD_PGHSWBORFD_WLRQ

_       _DOH[   _       _$&&2817,1*1(:<25.1< _
_       _WHG    _       _$&&2817,1*1(:<25.1< _
_       _MRH    _       _$&&2817,1*1(:<25.1< _
_       _PDW    _       _5(6($5&+_'$//$6 7;  _
_       _PD[    _       _5(6($5&+_'$//$6 7;  _

URZVQVHW    VHF
```

$ PDS VLGH MRLQ LV VLPLODU WR DQ LQQH
WKH PDSSHU DORQH QRWH WKDW WKH UH\
EHLGHQWLFDO

,Q JHQHUDO JHMRLDQWDYRHQWVDUH H[SHQVLYH
WKH HQWLUH FRQWHQWSDQEWKRWHQHQDUH JHW
WDO*HLYHQ WZSRQGEPHDS VLGH MRLQ ZLOO
WDSEOHDOHGWWKLEJHDUQ%WDEOHPHQVLRQV WW
WR PHGRXSHU7RUP WKLV W\SH RI MRLQ %ZDC
EURDGFDVW LW WR DOO QRGHV 1HZ[WWKZE
DQG WKHQ DFFHVV WKH UHG9HWKUQWKJLQ MRKF
KDVK WDEOH

7ZR GHPRQVWUDWH ZH¶OO (FU3HDQGG7 WWDZEO5
ZH FU(HDW3DV5D'Q GHSWBLG HPSBLG HPSBQ

```
(03 VSDUNSDUN&RQWHUWOOHOL]H
>
                DOH[
                WHG
                PDW
                PD[
                MRH
@
```

1H[W ZH'(F3U7HDDVWDHQ GHSWBLG GHSWBQDPH GHSWBORFDWLRQ @

```
'(37 VSDUNSDUN&RQWDLQHUOOHOL]H
>       $&&2817,1*1(: <25. 1<
        5(6($5&+ '$//$6 7;
        6$/(6 &+,&$*2 ,/
        23(5$7,216 %26721 0$
        0$5.(7,1* 6XQQ\YDOH &$
        62)7:$5( 6WDQIRUG &$
@
```

(03DQG37KDYH D FRPHSQWNWR ZH FDQ MRLQ DV IROORZV

```
!!! VRUWHG3MRLQ(37 FROOHFW
>
              DOH[   $&&2817,1*1(: <25. 1<
              WHG    $&&2817,1*1(: <25. 1<
              MRH    $&&2817,1*1(: <25. 1<
              PDW    5(6($5&+ '$//$6 7;
              PD[    5(6($5&+ '$//$6 7;
@
```

+RZ GRHV D PDS VLGH MRLQ R(S3LLVPDHHGDK DQ'G37LV D UHODWLYHO\ VPDOO GDWDQ3HZV '(37RQHSWBZGH ZLOO FUHDWH D EURDGFDV XVLQJ WKH FXWRRKDXEQH

```
EXLOG D GLFWLRQDU\ RI NH\ YDOXH
ZKHUH NH\ GHSWBLG
    YDOXH GHSWBQDPH GHSWBORFDWLRQ

GHIWRBKDVKBWGHSWB DVBOLVW
  KDVWBWDE^OH
  IRUGLQGHSWBDVBOLVW
    GHSWBLG> @
    GHSWBQDPHBORFG>W@LRQ
    KDVKBW>GHSWB@LGGHSWBQDPHBORFDWLRQ
  UHWXKDVKBWDEOH
 HQG GHI

GHSWBKDVKBWDRBKDVKBW(3EFROOHFW
```

$OWHUQDWLYHO\EXLOGWKHKDVKWDEOHX
FROOHFW$V0DSLFKUHWXUQVWKHSDLUV(3LQ
WRWKHPDVWHUDVDGLFWLRQDU\

```
GHSWBKDVKBWDEOHDOOHFW$V0DS
```

1RZXVHSQVJSDUN6SDUN&RQWH[WEDURDFGFDVWHG
WKHUHDGRQO\WDKUDLDKEDWDEWKHH6SDUNLFQG
DYDLODEOHIRUDOONLQGVRIWUDQVIRUP

```
VF VSDUNSDUN&RQWH[W
KDVKBWDEOHBEURDGFDVWHGVFEVRDHGFDVWWBKDVKBWDEOH
```

7RSHUIRUPWKHPDSVLGHMRLQLQWKH

```
GHSWBKDVKBWKDEVKHBWDEOHBEURDGFDVWHG
```

8VLQJWKHPDXSBFVWGRHMRGLQLQHGDVIROOR

```
H DV DQ HOHPHQW RI (03 5''
GHIPDSBVLGHBMRLQ
GHSWBLH> @
JHW KDVKBWDEOH IURP EURDGFDVWHG REMHFW
KDVKBWDEKDVKBWDEOHBEURDGFDVWHG
GHSWBQDPHBORFKDWLKBQV>GHSW@LG
UHWXUQ GHSWBQDPHBORFDWLRQ
HQG GHI
```

ZHFDQWKHQSHUIRUPDPSWKHDMRLQXVLQJDQ

```
MRLQHG03PDSPDSBVLGHBMRLQ
```

7KLVDOORZVIXOVHWWRKHQFGWPVHQXV3RQDQGEWRH
JRRGMRLQSHUIRUPDQFH

:LWKDPDSVLGHMRDSIZKQFXVRQXWRLWKH
URZR(0WKHEOH DQGUHWULHYHWKHSGLBQ
DQGHSWBORFDWLRQWKHEURDGFDVWHGXKDF

ZLOO EH H[HFXWHG FRQFXUUHQWO\ IRU HD
RI WKH KDVK WDEOH

7